

Nuclear Quantum Effects in Water and Aqueous Systems: Experiment, Theory, and Current Challenges

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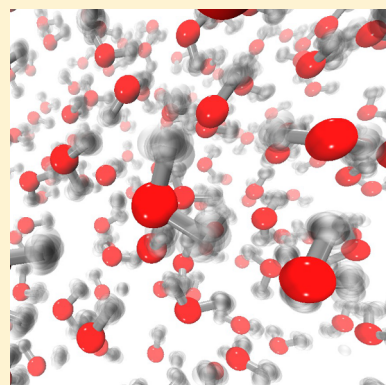
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ABSTRACT: Nuclear quantum effects influence the structure and dynamics of hydrogen-bonded systems, such as water, which impacts their observed properties with widely varying magnitudes. This review highlights the recent significant developments in the experiment, theory, and simulation of nuclear quantum effects in water. Novel experimental techniques, such as deep inelastic neutron scattering, now provide a detailed view of the role of nuclear quantum effects in water's properties. These have been combined with theoretical developments such as the introduction of the principle of competing quantum effects that allows the subtle interplay of water's quantum effects and their manifestation in experimental observables to be explained. We discuss how this principle has recently been used to explain the apparent dichotomy in water's isotope effects, which can range from very large to almost nonexistent depending on the property and conditions. We then review the latest major developments in simulation algorithms and theory that have enabled the efficient inclusion of nuclear quantum effects in molecular simulations, permitting their combination with on-the-fly evaluation of the potential energy surface using electronic structure theory. Finally, we identify current challenges and future opportunities in this area of research.



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Special Issue: Water - The Most Anomalous Liquid

Received: November 15, 2015

Published: April 6, 2016

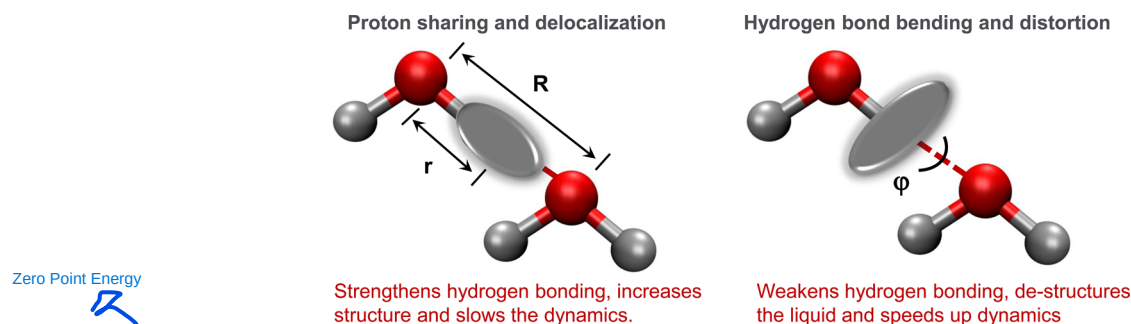


Figure 1. Competing quantum effects in the hydrogen bonding between two water molecules. There are two qualitatively different contributions to the vibrational ZPE. One is associated with the O–H stretch, shown on the left. The second contribution comes from the two bending vibrational modes: one in the plane of the water molecule, shown on the right, and the other perpendicular to the plane. As the distance, R , between the oxygen atoms decreases, the contribution of the stretch decreases, and that of the bend increases. Consequently, the two contributions strengthen and weaken the hydrogen bond, respectively.

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1. INTRODUCTION

Because of the low mass of the proton and the central role of hydrogen bonding, nuclear quantum effects (NQE), such as zero-point energy (ZPE) and tunneling, can play an important role in determining water's static and dynamical properties. Understanding these effects and developing methods to efficiently simulate them are thus vital to obtaining a chemically accurate description of water's properties from simulations and to elucidating experimentally observed static and kinetic isotope effects. Although aspects of NQEs in water have been reviewed previously,^{1–3} there have been significant recent developments such as an improved theoretical understanding of their role in modulating hydrogen-bond strengths, new algorithms for efficiently including them in simulations, and new experimental techniques for probing them.

In particular, the picture of competing quantum effects has provided significant insights into the interplay between different NQEs in the hydrogen bond.^{4–9} This concept is based on the observation that NQEs lead to an extension of the oxygen–hydrogen (O–H) covalent bond, allowing the protons to be more shared (delocalized) between hydrogen-bonded pairs of water molecules, as shown in the left panel of Figure 1. This effect acts to strengthen the hydrogen bond. However, quantum fluctuations also allow the protons to spread in the other directions, leading to distorted and, hence, weaker hydrogen bonds, as shown in the right panel of Figure 1. Which of these effects dominates is strongly determined by the distance between the oxygen atoms of the hydrogen-bonded water molecules, with short hydrogen bonds being made stronger upon the inclusion of NQEs whereas long ones are made weaker. This concept of “competing quantum effects” has been shown during the past decade to be a highly useful organizing principle for providing a much clearer understanding of the large variation in the magnitude of isotope effects observed in water.^{4,6–8}

This theoretical picture has been coupled with new algorithms that have greatly reduced the cost of performing simulations that include NQEs, in some cases yielding schemes with costs comparable to those of the corresponding classical

simulations.^{10–18} These algorithmic advances have allowed simulations of liquid water with NQEs to be performed where the molecular interactions are computed using on-the-fly electronic structure calculations^{7,19–24} and on potential energy surfaces fitted to high-level ab initio calculations.^{25–29} In addition, experimental techniques such as deep inelastic neutron scattering (DINS) are now able to probe the proton and oxygen momentum distributions and quantum kinetic energies, providing intriguing new observations of these quantum properties of the nuclei under thermodynamic conditions ranging from supercooled to supercritical water and environments ranging from the bulk to hydrophobic confinement.^{28,30–44}

Given the vast range of environments and aqueous systems in which NQEs can occur, this review cannot hope to be comprehensive. Hence, our focus in this review is to highlight some of the latest developments in the application of theory, modeling, and experiments to NQEs in water and to offer a perspective on how these developments will build into new discoveries in the future. First, we summarize experimental isotope effects observed in the static and dynamic properties of water with an emphasis on large and seemingly anomalous effects. Second, we discuss recent theoretical and simulation work that has allowed many of these effects to be elucidated. We then summarize methods for the efficient and accurate inclusion of NQEs in simulations to obtain static and dynamical properties of water. Finally, we summarize the ongoing challenges and opportunities that remain in understanding and modeling NQEs in water.

2. EXPERIMENTAL OBSERVATIONS OF NUCLEAR QUANTUM EFFECTS IN WATER

In this section, we provide a brief overview of various micro- and macroscopic properties of water that exhibit notable experimental isotope effects. In particular, we emphasize that, although many experiments suggest that NQEs act to weaken the hydrogen bond, leading to a less structured liquid and a more mobile hydrogen-bonded network in H₂O than in the less quantum mechanical D₂O, this is not always the case and can depend on both the conditions and the property of interest. Where appropriate in this section, we mention relevant simulation work used to analyze the experiments. However, the bulk of the simulation work and the theoretical origins of water's seemingly ambivalent response to the inclusion of NQEs is covered in more detail in section 3.

Table 1. Experimentally Observed Isotope Effects in Water^a

property	H ₂ O	D ₂ O	T ₂ O	H ₂ ¹⁸ O	ref(s)
• bond energies (kJ/mol) (gas phase)	458.9	466.4 (1.6%)			49
• dipole moment (D) (gas phase)	1.855	1.855			50
vibrational frequency (cm ⁻¹) (gas phase)	3657.1	2671.6	2237.2	3649.7	51, 52
	1594.7	1178.4	995.4	1588.3	
	3755.9	2787.7	2366.6	3741.6	
water dimer dissociation energy (kJ/mol) (gas phase) 10 K	13.22	14.88 (12.7%)			53–55
• melting point, T _m (K) (at 1 atm)	273.15	276.97 (1.40%)	277.64 (1.64%)	273.46 (0.11%)	52, 56, 57
• temperature of maximum density (TMD) (K)	277.13	284.34 (2.60%)	286.55 (3.40%)	277.36 (0.08%)	58, 59
critical temperature (K)	647.10	643.85 (−0.50%)	641.66 (−0.84%)		52, 56
molar density (mol/L)	55.35	55.14 (−0.38%)	55.08 (−0.49%)	55.42 (0.13%)	56, 60
molar density at the TMD (mol/L)	55.52	55.22 (−0.53%)	55.17 (−0.63%)	55.59 (0.13%)	56, 59
liquid/vapor surface tension (N/m)	0.07198	0.07187 (−0.15%)			56
• specific heat capacity, C _v [J/(K mol)]	74.54	84.42 (13.2%)			56, 61
translational diffusion (Å ² /ps)	0.230	0.177 (−23.0%)		0.219 (−6.1%)	62–64
rotational diffusion (rad ² /ps)	0.104	0.086 (−17%)			65
• dielectric relaxation time at 20 °C (ps)	9.55	12.3 (29%)			66
• viscosity (mPa s)	0.8904	1.0966 (23.2%)		0.9381 (5.4%)	56, 59
acidity pH/pD	7.00	7.43	7.6		56, 57

^aValues given are at 25 °C unless otherwise stated, and values in parentheses are percentage shifts relative to H₂O.

Isotope effects for a range of properties are summarized in Table 1 for the exchange of H for deuterium (D) or tritium (T), or of ¹⁶O for ¹⁸O. Whereas H₂O and D₂O data are available for many of water's properties in its numerous chemical phases, much more limited data sets are available for H₂¹⁸O and T₂O. We have included values for H₂¹⁸O where available, because it has the same molecular mass as D₂O, thus allowing one to compare the influence of the molecular mass, which changes upon ¹⁶O/¹⁸O substitution, and effects due to the moment of inertia and proton ZPE that change upon H/D substitution. At the outset, it is interesting to note that water's seeming ambivalence to quantum effects for some properties is reflected in the fact that prokaryotic organisms such as bacteria can tolerate pure-D₂O conditions,⁴⁵ whereas deuterating about 30% of plants' and mammals' water can be lethal.^{46–48}

2.1. Energy Scales of Quantum Effects in Water

Isotope effects in water range from those observed under ambient conditions, which can often be attributed to the ZPEs in the O–H stretch or other degrees of freedom, to those at high pressures or low temperatures, where tunneling and proton delocalization become particularly important.^{67–74} The ZPE of the O–H stretch is equivalent to a ~2000 K rise in temperature along that coordinate and is modulated by the local hydrogen-bonded structure of water, for example, in different phases, at interfaces, or in the presence of solutes. At high pressures and/or low temperatures, the uncertainty in the proton's position becomes comparable to the distance between the minima in the potential energy surface. This can lead to proton tunneling and extensive delocalization that are strongly coupled to changes in the distances between oxygen atoms.^{4,20,75}

NQEs manifest as isotope effects upon substitution of H for D or T or of ¹⁶O for ¹⁸O. Although it is common to view deuterated water as a “classical” analogue of water, it is important to note that, even though D₂O and T₂O have smaller NQEs, they are not classical species. The vibrational frequencies in Table 1 allow one to estimate that the ZPE in liquid water is ~21 kJ/mol per O–H stretch and that the O–D and O–T stretches still exhibit ZPEs of ~15 and ~12 kJ/mol,

respectively, that is, still ~71% and ~58% of that in H₂O. One can also estimate the total harmonic contribution to the zero-point energy in the gas phase to be ~54 kJ/mol for H₂O and ~40 kJ/mol for D₂O. These estimates give an idea of the energy scale of NQEs at room temperature, as compared to the classical contribution to the (intramolecular) kinetic energy of 3.7 kJ/mol at 300 K. In addition, although ¹⁶O/¹⁸O substitution generally gives rise to small isotope effects, such effects are widely used in atmospheric isotope fractionation studies that form crucial inputs to climate modeling.^{76,77}

2.2. Thermodynamic Properties

Gas-phase properties of the water molecule are listed in Table 1. The covalent-bond energies exhibit a quantum effect, with the O–D bond being about 1.6% stronger than the O–H bond. This amounts to a difference of 7.5 kJ/mol, which is largely explained by the difference of 5.8 kJ/mol in the O–H and O–D ZPEs. D₂O has essentially the same gas-phase dipole moment as H₂O, which is perhaps not unexpected given that their bond lengths and angles are almost identical (less than 0.05% difference in both cases).⁷⁸ However, D₂O has a dimer dissociation energy that is about 12.7% larger than that of H₂O, suggesting that D forms a substantially stronger hydrogen bond than H in the gas phase. In addition, the relative stabilities of the low-temperature structures of the water hexamer have also been shown to be highly sensitive to NQEs.^{79–81}

+ dissociation energy = stronger HB

A number of observations suggest that the stronger hydrogen bond observed for D₂O in the gas phase is also present in the liquid. For example, the melting point is increased by 3.82 K upon deuteration, which is indicative of the idea that D forms a stronger hydrogen bond that leads to an increase in the stability of ice over the liquid. The temperature of maximum density (TMD) is increased by 7.21 K upon deuteration, and hence, the difference between the melting point and the TMD increases from 3.73 to 7.37 K, indicating that a universal scaling law applied to the TMD is insufficient to account for these isotope effects. The effects of NQEs and isotope substitution on the melting point,^{82–84} TMD,^{6,83} and phase diagram of water⁸⁵ have been quite well reproduced theoretically through simulations using the q-TIP4P/F water model. The heat

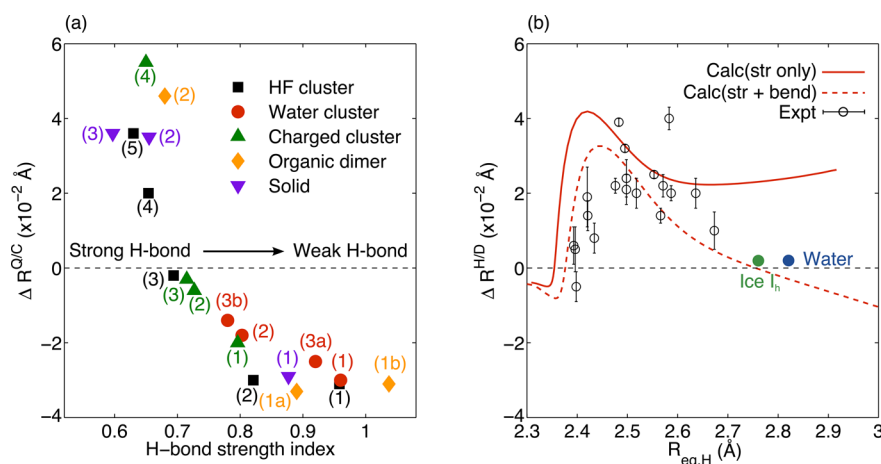


Figure 2. Secondary geometric NQEs in H-bonds. (a) Differences between the heavy-atom distances (R) in a H-bond from MD simulations (with classical nuclei) and PIMD simulations (with quantum nuclei) ($R^C - R^Q$) vs H-bond strength (defined as the ratio of the X–H stretching frequency in the H-bonded cluster to that in the free monomer). A positive value means that the H-bond is shorter when NQEs are incorporated. For the HF clusters, labels 1–5 refer to the H-bonds in the dimer to the hexamer, respectively. For the water clusters, labels 1, 2, and 3a (3b) refer to the H-bonds in the dimer, the H-bonds in the pentamer, and the long (short) H-bonds in the octamer. For the charged clusters, labels 1–4 refer to $H_3O_3^+$, $H_3O_4^+$, $H_7O_4^-$, and $H_2H_5^-$, respectively. For the organic dimers, labels 1a, 1b, and 2 refer to the red-shifted and blue-shifted H-bonds in formamide and the H-bond in formic acid, respectively. For the solids, labels 1–3 refer to the H-bonds in HCl, HF, and squaric acid, respectively. Panel a adapted with permission from ref 9. Copyright 2011 National Academy of Sciences. (b) Dependence of the secondary geometric isotope effect on the O–O distance R . This refers to the change in the O–O distance upon H/D substitution ($R^D - R^H$). Experimental data for a range of different compounds are compared to the magnitudes calculated from the simple diabatic state model.^{4,5} The solid and dashed curves are the predictions when only the ZPE of the O–H stretch is taken into account and when the O–H bending modes are also taken into account, respectively. Note that, for weak hydrogen bonds, the effect changes sign, that is, weak hydrogen bonds get stronger upon H/D substitution. Panel b adapted with permission from ref 4. Copyright 2014 AIP Publishing. This figure illustrates how the overall NQE in water is relatively small as a result of competing quantum effects.

capacity of the liquid is also considerably larger for D_2O than for H_2O at a given temperature (see Table 1), and recent work by Vega et al. highlighted that this is a reflection of the importance of NQEs in water even under ambient conditions.⁸⁶ Comparison of the molar densities, both at the temperature of maximum density and at 298 K, reveals a small expansion for D_2O and T_2O and a small contraction for $H_2^{18}O$. Although it might be tempting to interpret these changes in density as suggesting stronger hydrogen bonding in the more dense H_2O , one should be careful given that, because of the directional nature of hydrogen bonds, **stronger hydrogen bonding can lead to decreased density** [e.g., as observed in simulations based on descriptions of water using Kohn–Sham density functional theory (DFT) with the generalized gradient approximation (GGA)⁸⁷]. The pH of liquid water is also increased from 7.0 to 7.4 upon deuteration and to 7.6 upon tritiation.^{56,57} These data are consistent with a quasiharmonic scaling that predicts that the pK_a varies as the inverse square root of the particle mass.¹⁰ Given this relation, one can extrapolate to the classical (infinite-mass) limit, which implies that the pH of classical water should be ~ 8.5 , that is, a ~ 30 -fold change in the acid dissociation constant.

All of these observations are broadly consistent with a picture that **NQEs weaken the hydrogen bond in liquid water** and, therefore, **that the hydrogen bond is strengthened upon deuteration**. However, **other properties are not consistent with such an interpretation**. For example, the critical temperatures of D_2O and T_2O are lower than that of H_2O , suggesting that, at these higher temperatures, NQEs act to strengthen hydrogen bonds—in contrast to the observed behavior at lower temperatures. Comparison of the liquid/vapor surface tension also indicates a seemingly anomalously smaller value for D_2O relative to H_2O .

Upon H/D substitution, the volumes of liquid water and of ice I_h increase by 0.1%. This increase is anomalous, as many liquids and solids exhibit a volume decrease upon substitution with a heavy isotope.⁸⁸ Furthermore, the magnitude of the change for water is anomalously small for a hydrogen-bonded compound. **Figure 2b shows experimental data for the change in O–O bond length upon H/D substitution for a wide range of compounds. This is referred to as the secondary geometric isotope effect. The primary geometric isotope effect refers to changes in the O–H bond lengths. Figure 2b also compares the experimental results with predictions based on the simple diabatic state model discussed in section 3.4.⁴ Of particular note is that, for liquid water and ice I_h , the effect (~ 0.001 Å) is relatively small, highlighting that the competing quantum effects in water under these conditions almost exactly cancel. Similar results and conclusions have been obtained from path-integral molecular dynamics (PIMD) simulations, as discussed below.**

2.3. Spectroscopic Properties

X-ray spectroscopies^{89,90} that probe the local structure in liquid water have provided further evidence that **D_2O makes stronger hydrogen bonds than H_2O at room temperature**. X-ray Raman scattering (XRS)⁸⁹ experiments have been used to compare the structures of H_2O and D_2O at different temperatures, revealing that the structure of H_2O at a given temperature appears similar to that of D_2O at a higher temperature. XRS, X-ray absorption spectroscopy (XAS),^{91–93} and X-ray emission spectroscopy (XES)^{94–96} have been used to propose and probe the balance of two structural classes of hydrogen bonds, tetrahedral and asymmetric, in liquid water. Comparison of the XRS⁸⁹ spectroscopic features assigned to the asymmetric species indicated that the asymmetric hydrogen bonds in H_2O show

enhanced asymmetry compared to those in D_2O . This asymmetry is present even when the effect of an offset in temperature is taken into account. In addition, upon comparison of the vibrationally resolved spectra for H_2O , D_2O , and HDO ⁹⁰ using resonantly excited XES in which asymmetrically hydrogen-bonded molecules (with one strong and one weak/broken hydrogen bond in the HDO liquid) were selectively excited, a strong preference was observed for OD as the strong hydrogen bond and OH as the weak/broken hydrogen bond. A ratio as high as 1.5 (± 0.2):1 was found compared to the 1:1 ratio that would be expected for classical water.⁹⁰ Hence, the common practice in femtosecond pump–probe vibrational spectroscopies, where dilute HDO in either H_2O or D_2O is probed on the assumption that the OH and OD stretches probe the same hydrogen-bond situations with equal statistical weight, is challenged when NQEs are considered.

The concept that D_2O is a more structured liquid than H_2O at ambient temperatures is also supported by X-ray, γ -ray, and neutron scattering experiments.^{97–99} These studies indicate a general broadening of structural features in going from heavy to light water, where the radial distribution functions (RDFs) for liquid D_2O appear to be slightly more structured, characterized by more pronounced peaks in the O–O RDF, as shown in Figure 3, than those for liquid H_2O , with the O–H covalent bond being longer than the O–D bond by as much as $\sim 3\%$.⁹⁸ In turn, the hydrogen bond is $\sim 4\%$ shorter, the first intermolecular OH peak is more asymmetric, and the first

HH intermolecular distance is $\sim 2\%$ longer in H_2O compared to D_2O .⁹⁸ The enhanced hydrogen-bonding asymmetry in H_2O compared to D_2O deduced from XRS⁸⁹ is thus further supported by the combined X-ray and neutron scattering data.⁹⁸ However, other neutron scattering results using oxygen substitution and a different analysis technique¹⁰⁰ suggested the change in the length of the covalent bond upon deuteration to be significantly smaller at about 0.5%.^{101,102} Hence, there still remain significant uncertainties in the experimental difference between the RDFs for light and heavy water.

2.4. Dynamical Properties

Data on how isotopic substitution affects the dynamical properties of water are also listed in Table 1. In particular, we report information on the translational diffusion, rotational diffusion, dielectric relaxation time, and (dynamic) viscosity. These results are all also consistent with D having a stronger hydrogen bond at about 300 K, with the mobility of D_2O being about 20–30% lower than that of H_2O . The 23% rise in the viscosity upon deuteration is accompanied by a 23% decrease in diffusion, consistent with Stokes–Einstein behavior. It is important to note that isotope substitution gives rise to greater proportional changes in the dynamical properties than those observed for most static properties. This is partially because dynamical events are driven by rare fluctuations in the hydrogen-bond network that are altered much more by quantum effects than the average values of the properties. However, it is also because, even in the classical limit, changing H to D results in a change in the dynamical properties, unlike static equilibrium properties, because of the increase in the mass. This trivial contribution, which does not arise from NQEs, results in a slowing of dynamical properties upon substituting H for D. This slowing ranges from a factor of $(m_{\text{H}}/m_{\text{D}})^{1/2} = 0.707$ if the dynamical process arises from pure ballistic motion of the substituted atom to $(m_{\text{H}_2\text{O}}/m_{\text{D}_2\text{O}})^{1/2} = 0.949$ if the process involves ballistic motion of the center of mass of the water molecule. Notably, NMR experiments¹⁰³ show that, at higher temperatures (~ 573 K), the translational diffusion constants for D_2O and H_2O undergo an apparent crossover (i.e., their ratio drops below the expected change of 0.949).

D_2O has stronger HB, higher viscosity and lower mobility (diffusion) than H_2O .

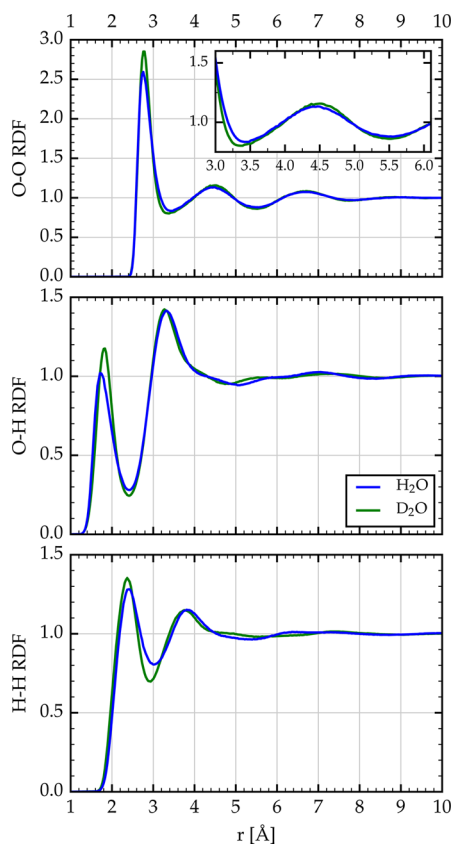


Figure 3. Radial distribution functions of liquid water for H_2O (blue line) and D_2O (green line) at 296 K obtained from refinement of neutron and X-ray scattering experiments. D_2O has a higher first peak in the O–O radial distribution function, suggesting that it forms a more structured liquid. Figure created from data in ref 98.

2.5. Tunneling and Proton Delocalization Effects

At low temperatures or when water molecules are forced into close proximity (such as at high pressure or at interfaces), more pronounced quantum effects can occur, arising from proton tunneling and delocalization. In particular, experimental and theoretical studies on high-pressure ice have shown quantum tunneling of protons in ice VII,^{67,71} whereas higher-pressure ices such as ice X exhibit symmetric hydrogen bonds, where the protons are located halfway between two water molecules.^{71,104,105} The adsorption of water on solid surfaces is an area that has received considerable attention from the surface-science community,^{106–108} and on some surfaces, water molecules can be forced into close proximity because of their interaction with the substrate. The shortened distances between neighboring molecules can then lead to quantum proton delocalization,¹⁰⁹ as shown, for example, in scanning tunneling microscopy (STM) experiments of water clusters on $\text{Cu}(110)$ ¹¹⁰ and Au-supported $\text{NaCl}(001)$ films, where it was possible to directly probe concerted proton tunneling in water tetramers.¹¹¹

2.6. Solvation Properties

It is also worth briefly commenting on the effect that NQEs have on changing the solvation behavior of species. For the solvation of large biomolecules,^{112,113} small molecules,^{114–116} and ions,¹¹⁷ small changes have been observed in the dynamics, solvation free energy, and structure surrounding them upon the inclusion of NQEs. Indeed, in experiments on gas-phase HOD hydrogen-bonded complexes with fluoride and iodide ions, the former ion binds preferentially to OH and the latter to OD.¹¹⁷ This inversion of behavior for ions that strongly and weakly bind water further suggests a delicate cancellation between different NQEs. A recent simulation study suggested that there is an absence of notable NQEs in the solvation of monovalent ions, such as Li⁺ and F[−], because of competing quantum effects (see following section),¹¹⁸ although another recent study treating the water molecules as rigid entities and, hence, neglecting competing quantum effects observed the hydrogen-bond dynamics of water around F[−] to be nearly 40% faster than observed in classical simulations.¹¹⁹

When light species, such as solvated protons,^{2,120–124} hydroxide,^{125,126} hydrogen atoms,^{127,128} and electrons,^{129–137} which themselves exhibit notable NQEs, are solvated in water, larger effects are often observed. For example, NQEs reduce the free energy barrier to bulk proton transfer by a factor of ~3.^{138,139} Experimentally, proton diffusion has been observed to be slowed by a factor of ~1.5 upon deuteration.¹⁴⁰ Hence, the excess proton in water diffuses ~4.1 times faster than water itself, whereas D⁺ diffuses faster by only a factor of ~3.6 in D₂O. Diffusion of hydrogen atom and muonium (Mu), an isotope of hydrogen with approximately one-ninth the mass, has also been studied in liquid water and hexagonal ice by both experiment^{141–143} and simulation.^{127,128} In room-temperature water, little change in the diffusion constant is observed upon changing the atom from Mu to H. This has been explained by considering the effective swelling of the water cavity that holds the Mu atom due to the zero-point pressure exerted by the quantum particle on the water.¹²⁷ This behavior leads to a turnover such that Mu diffuses faster than H at temperatures above ~300 K whereas, below that temperature, both in the liquid and in hexagonal ice, H diffuses faster. At very low temperatures, where tunneling dominates, Mu is again observed to diffuse faster than H.¹⁴¹

2.7. Absolute Measures of Nuclear Quantum Effects

Although the changes in the aforementioned properties upon isotope substitution provide an indication of the role of NQEs, even tritiated water is far from the classical limit, making it hard to assess their total contribution. Properties for which the **classical limit can be predicted exactly** are therefore particularly interesting, as they **give an absolute measure of NQEs**.

For instance, the logarithm of the equilibrium isotope fractionation of hydrogen and deuterium between liquid water and its vapor would be zero if the nuclei were classical particles.⁸ **These water fractionation ratios are used to provide a means of monitoring spatial and historical variations in atmospheric temperatures.**⁷⁷ For example, fractionation ratio measurements from Antarctic ice cores are used to provide information about global temperatures over the past 300000 years.¹⁴⁴ **Figure 4 shows the fractionation ratio plotted as $10^3 \ln(\alpha_{l-v})$, with α_{l-v} defined as**

$$\alpha_{l-v} = \frac{(x_{D,l}/x_{H,l})}{(x_{D,v}/x_{H,v})} = e^{-\Delta A/k_B T}$$

Fornecer a fração de água líquida e vapor existentes com base na temperatura

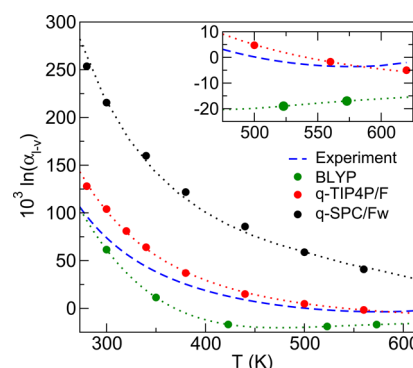
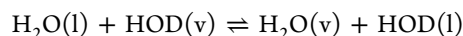


Figure 4. Liquid–vapor isotope fractionation ratio versus temperature from experiment²⁷⁹ and theory.^{8,20} Variation in the temperature corresponds to moving along the liquid–vapor coexistence curve. In the temperature range shown (up to 600 K), the density of the liquid phase changes by a factor of ~3. Hence, there is a significant variation in the O–O distance and the strength of the hydrogen bonding. At about 500 K, the experiments show an inversion in the fractionation ratio, indicating that D is preferentially found in the gas phase above this temperature.

where x_Z is the mole fraction of isotope Z, l denotes the liquid phase, and v denotes the vapor phase. ΔA is the Helmholtz free energy corresponding to the process



Hence, a **positive value** ($10^3 \ln \alpha > 0$) corresponds to a **preference for H to reside in the gas phase compared to D**, and a **negative value** corresponds to a **preference for it to reside in the liquid phase**. If hydrogen bonds formed by D in the liquid are stronger than those formed by H, then one would expect an excess of D in the liquid, where hydrogen bonds can be formed, compared to in the gas, where they are not.

HB do D são mais fortes que H, então D seria mais em excesso nos líquidos

Isotope fractionation ratios can also be shown to be directly related to the quantum kinetic energy difference between the isotopes.^{8,10,145,146}

As shown in **Figure 4**, at room temperature, experimentally (blue line), $10^3 \ln \alpha$ is positive, indicating a preference for D to reside in the liquid, which suggests that it forms a stronger hydrogen bond. However, as the temperature is raised above 493 K, the sign of $10^3 \ln \alpha$ inverts, indicating that H is preferentially found in the liquid phase. This inversion, which can be explained by the principle of competing quantum effects, provides another example that water exhibits a dichotomy in its NQEs under different conditions. As we discuss in **section 3.2**, the ability to model this effect is highly sensitive to the anharmonicity of the O–H stretch in the potential energy surface used.

DINS experiments can be used to provide measurements of the quantum kinetic energy and momentum distributions of protons, deuterons, and more recently heavier nuclei.¹⁴⁷ In the classical limit, the total kinetic energy of a nucleus reduces to its classical equipartition value ($3k_B T/2$), and the momentum distribution is simply the Maxwell–Boltzmann distribution. However, because of the quantum nature of nuclei, both quantities vary depending on the environment and can be related to the shape of the potential experienced by the nucleus.^{16,37,44,67,148–151} DINS measurements thus rely on the quantum nature of nuclei to extract valuable information on the structure and behavior of light atoms in the condensed phase. These experiments have now been performed extensively on

Classicamente:

Energia cinética do núcleo = equipartition $3k_B T/2$

distribuição de momentum = Maxwell-Boltzmann distribution

Na quântica, elas dependem da forma do potencial do núcleo

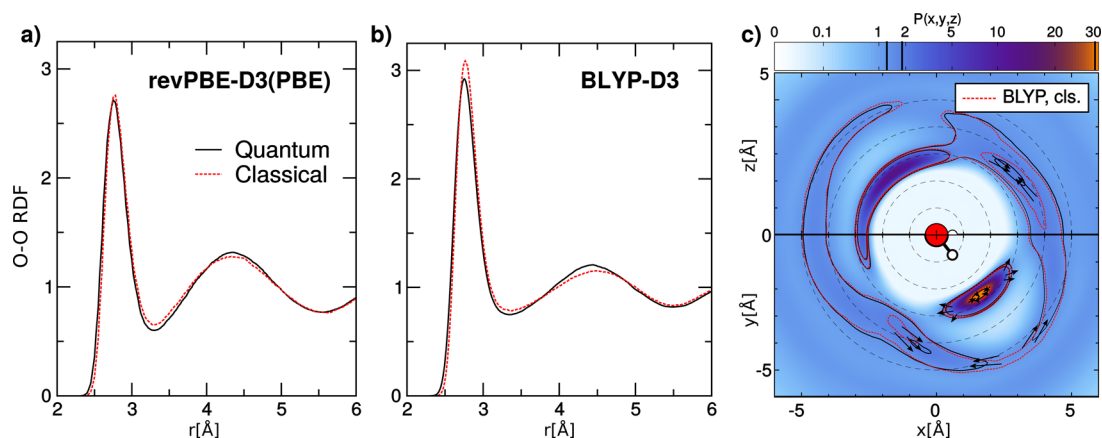


Figure 5. Radial distribution functions obtained from ab initio molecular dynamics simulations of liquid water at 300 K, including nuclear quantum effects (black line) or with classical nuclei (red line). (a) Results for the revPBE functional with the PBE D3 vdW corrections [revPBE-D3(PBE)] obtained from a 100-ps simulation using ab initio ring-polymer contraction.¹⁵⁵ (b) Results obtained for BLYP with D3 vdW corrections (BLYP-D3) using a 150-ps classical parallel tempering run from which four independent quantum trajectories of 25 ps each using six beads and PIGLET thermostating were launched (for a total of 100 ps of PIMD following the classical equilibration).¹⁵⁶ (c) Slices of the 3D O–O distribution function along the plane of the molecule (xy) and the perpendicular plane (xz) for BLYP-D3; the color scale and the black contours correspond to the simulations with quantum nuclei, and the red dashed contours correspond to the same density values but for classical simulations. Arrows highlight the changes in probability distribution triggered by including NQEs. Panel a adapted with permission from ref 155. Copyright 2016 AIP Publishing. Panels b and c adapted with permission from ref 156. Copyright 2016 American Chemical Society.

bulk water at different points of its phase diagram,^{30–32,34–38,40,151} suggesting that protons in confined water exhibit an anomalous excess of kinetic energy.³⁹

In addition, by measuring the different components of the kinetic energy tensor in different phases, DINS provides perhaps the most direct experimental probe of competing quantum effects. Specifically, Romanelli et al.³² found that the principal values of this tensor change in opposite directions when heavy-water ice melts. This significantly reduces the net quantum effect on the kinetic energy change and can be related to the relatively small net quantum effect on the melting temperature. These experimental results agree qualitatively with PIMD simulations using potential energy surfaces from DFT.³²

3. SIMULATIONS AND THEORY OF NUCLEAR QUANTUM EFFECTS IN WATER

In this section, we summarize some recent insights that have been obtained from simulations of water. Increases in computational power and improved algorithms to include NQEs in simulations (discussed in section 4) have led to a major surge of activity, including applications to systems ranging from bulk liquid water to water at surfaces. This has also enabled various comparisons with experiment to be made and provided insight into properties that are not directly observable by experiment. The ability to perform such calculations with different computational approaches has also made it possible to assess the reliability of different models.

3.1. Structure of Bulk Liquid Water

Early path-integral simulations of liquid water using empirical potentials employed rigid water molecules or described the O–H stretch using a purely harmonic potential. These studies appeared to show a consistent picture that NQEs weaken the hydrogen-bond network in water, leading to a less structured (lower and broader) first peak in the O–O RDF,^{152–154} which is broadly consistent with the experimental refinement of neutron and X-ray scattering data shown in Figure 3. To produce an equivalent change in the RDF by changing the

temperature in a classical simulation required an increase of about 35–40 K.

However, a study using a flexible empirical water model that included anharmonic terms in the O–H stretch revealed that this was an incomplete picture.⁶ Upon inclusion of the anharmonicity, the ZPE allows the proton involved in the hydrogen bond to extend in the direction of the acceptor oxygen to form a stronger hydrogen bond. This counterbalances the weakening effect, resulting in a smaller total impact of NQEs. These competing quantum effects, first observed in water, have since been extended to explain a much wider range of hydrogen-bonded systems using ab initio PIMD⁹ (which exactly includes NQEs on static properties for a given potential energy surface, as discussed in section 4). This work revealed a general trend that weak hydrogen bonds are weakened by NQEs whereas strong hydrogen bonds are strengthened by NQEs (Figure 2a).⁹ Behind this trend is the competition between the NQEs in librational modes and the NQEs in stretching modes, where the former dominate in relatively weak hydrogen bonds whereas the latter dominate in relatively strong hydrogen bonds.

The extent of this cancellation of NQEs in water has remained a matter of debate, however, because it depends sensitively on the electronic structure approach or force field used to model the water interaction—particularly in terms of the anharmonicity of the O–H coordinate. For example, an earlier study using DFT also hinted at competing quantum effects,⁷ but a subsequent simulation using the same exchange-correlation functional suggested that this was not the case.¹⁹ Because of the pioneering nature of these early path-integral calculations, the total time run in each of these simulations was less than 20 ps (including equilibration). Hence, statistical error bars are likely to be an issue in detecting the subtle differences that occur in the O–O RDF upon the inclusion of NQEs. However, because of the recent introduction of algorithms to efficiently include NQEs^{16,155} and current computational power, it has now become possible to perform simulations long enough to exhaustively sample the O–O RDF using ab

initio PIMD simulations. Figure 5 shows the results obtained from two recent studies using two different GGA functionals at 300 K. The quantum and classical nuclear RDFs shown in Figure 5a were each obtained from 100 ps of dynamics using the recently introduced ab initio ring-polymer contraction scheme¹⁵⁵ (see section 4.1.4), whereas those in Figure 5b were obtained from 150 ps of classical parallel tempering followed by 100 ps of simulation using the PIGLET^{16,156} algorithm (see section 4.1.3). In both cases, the qualitative changes to the RDF are consistent. Specifically, in the first coordination shell, the rise in the first peak occurs at a slightly shorter distance, which is complemented by a lowering of the first O–O peak height (integration of the quantum and classical RDFs to obtain the coordination number in the first shell indicates that, in both cases, the coordination number is essentially unchanged). The minimum following the first peak of the RDF is deeper in the quantum case for both functionals, followed by a slightly higher second peak, which suggests increased structure in the second coordination shell. Inspection of the slices of the 3D distribution function in Figure 5c for the BLYP exchange-correlation functional with D3 van der Waals (vdW) corrections allows one to pinpoint the origin of the increased structure of the second peak: **Interstitial** configurations, corresponding to distorted H-bond geometries, become less abundant in the presence of NQEs, with an increase in density in the region at about 4 Å. This destabilization of interstitial defects can explain the counterintuitive structure-enhancing effect of quantum delocalization. However, despite the consistency of these two GGA functionals as to the influence of NQEs on the structure of liquid water, more accurate descriptions of its electronic structure might alter this balance.

Although there is undoubtedly a degree of cancellation of different quantum effects on the O–O RDF, quantum effects on individual H-bond coordinates can be dramatic. The fluctuations of a proton along the covalent bond direction increase almost 10-fold relative to those in a classical simulation. It was observed that, as a consequence of quantum fluctuations, protons in liquid water experience transient excursions bringing them closer to the acceptor than the donor oxygen atom.^{20,21} Analysis of the electron densities (by means of maximally localized Wannier functions)¹⁵⁷ showed that these extreme quantum fluctuations are stabilized by considerable electron density rearrangements. However, the number of these excursions in liquid water remains a matter of debate because different DFT exchange-correlation functionals predict that these events occur for between 0.5% and 0.05% of the protons in water at a given time.^{20,21}

Quantum fluctuations along the hydrogen-bond direction can also favor genuine autoprotolysis and transition to delocalized-proton superionic states. PIMD simulations of liquid water at 10 GPa and 750 K, where the density is 65% higher than that of the ambient liquid, also show frequent proton exchanges between water molecules when NQEs are included that are not observed when the nuclei are treated classically.²⁰⁹ A similar interplay of pressure-induced delocalization and nuclear quantum fluctuations has been shown in action in exotic phases of ice at low temperature.^{67,158}

3.2. Isotope Fractionation as a Probe of Nuclear Quantum Effects

To try to clarify the nature and impact of the concept of competing quantum effects, a number of recent theoretical studies have concentrated on the ability of models to accurately

obtain the H/D isotope fractionation ratio between liquid water and its vapor. As discussed in section 2.7, the logarithm of this ratio would be zero in a classical simulation and is thus a very direct probe of NQEs because it is known very accurately under NVT conditions from experiments.^{8,10,20} The isotope fractionation ratio can be shown to relate directly to the quantum kinetic energy of the protons in water. Hence, by decomposing the quantum kinetic energy in a molecular reference frame,^{8,148,159} one can observe the contributions of its different components. The different components of the kinetic energy of heavy water near its melting point were obtained from DINS experiments, and ab initio PIMD results using the BLYP functional showed good agreement with the experimental data.³²

Based on this line of thinking, it was possible to show using PIMD simulations that the anharmonic flexible q-TIP4P/F model,⁶ which was the model originally used to suggest the existence of competing quantum effects in water, is able to accurately obtain the correct fractionation ratio over a wide range of temperatures,⁸ as shown in Figure 4 (red line). Moreover, it even accurately reproduced the experimental temperature where the fractionation ratio inverts, indicating that temperature affects the two competing quantum effects differently and, hence, can change their balance. This is in contrast to rigid and harmonic models, such as the harmonic q-SPC/Fw potential in Figure 4 (black line), which considerably overestimate the extent of fractionation compared to experiment and give no inversion point. By projection of the kinetic energy into the internal molecular coordinates, it was shown that the component of the fractionation arising from the O–H stretch undergoes a large cancellation (~90%) with the contribution of the in-plane and orthogonal librations. This example highlights that one must be careful when simulating quantum effects with force fields, because many force fields are parametrized to properties of chemical bonds near their equilibrium bond lengths. However, because of the large amount of ZPE, a quantum particle can explore a much larger part of the potential energy surface, where the force field might not be accurate. For example, the TTM3-F model, which was fit to high-level ab initio calculations,²⁶ shows a qualitatively incorrect inverted fractionation behavior at room temperature. This arises because the slope of the TTM3-F dipole moment surface at large O–H bond extensions increases the charge on the H too much upon hydrogen bonding.⁸ This leads to the dominance of hydrogen-bond strengthening when quantum effects are included and, hence, the incorrect inverted fractionation behavior at room temperature. However, the more extreme O–H bond lengths that lead to this failure are extremely rare in classical simulations, which lack the ~21 kJ/mol ZPE in that coordinate, and hence TTM3-F fares much better in classical simulations.

Recent developments in methods to efficiently include NQEs¹⁶ and simulate isotope effects^{10,146,160,161} have also allowed the assessment of these effects in properties such as the fractionation ratio when combined with an on-the-fly evaluation of the potential energy surface using a variety of DFT exchange-correlation functionals. Because treating competing quantum effects requires an accurate description of a much wider range of the potential energy surface, it is a sensitive probe as to whether an electronic structure method is able to correctly reproduce the behavior of hydrogen-bonded systems. For example, recent work studied the isotope fractionation of water with ab initio path-integral simulations²⁰ to assess the role of

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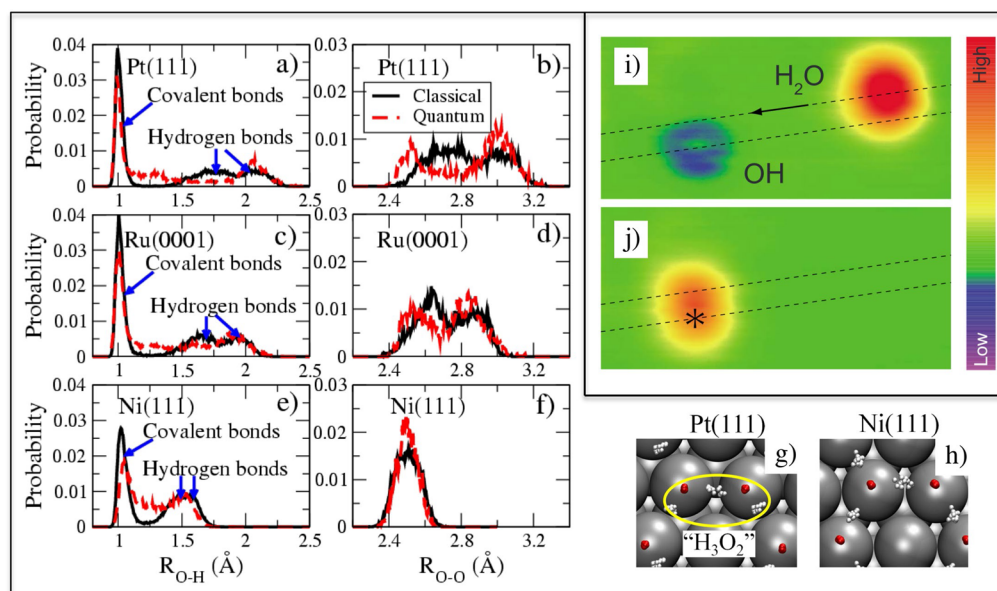


Figure 6. Evidence for strong quantum effects for water adsorbed on metal surfaces. Left: Simulation results for selected structural properties of extended water–hydroxyl overlayers on metal surfaces. Probability distributions of (a,c,e) O–H and (b,d,f) O–O distances on (a,b) Pt, (c,d) Ru, and (e,f) Ni at 160 K, as obtained from MD with classical nuclei (labeled classical, black solid lines) and PIMD (labeled quantum, red dashed lines). (g,h) Snapshots for typical spatial configurations of the overlayer on (g) Pt and (h) Ni obtained from PIMD (with every atom represented by 16 bead. On Pt and Ru at any given snapshot, one proton is equally shared by two of the O atoms, yielding an intermediate HO–H–OH-like complex. On Ni at any given snapshot, several protons can simultaneously be shared between the oxygens. (i,j) Scanning tunneling microscopy images (i) before and (j) after the formation of a water–hydroxyl dimer on Cu(110). The symmetric nature of the dimer in panel j implies that the hydrogen between the two oxygens in this structure is shared symmetrically. Panels a–h adapted with permission from ref 109. Copyright 2010 American Physical Society. Panels i and j reproduced with permission from ref 110. Copyright 2010 American Physical Society.

including exact exchange and vdW corrections in the PBE and BLYP functionals. The PBE functional was shown to overestimate the contribution of O–H stretching and predicted inverted isotope fractionation with respect to experiment at room temperature ($10^3 \ln \alpha = -17$), whereas the BLYP functional ($10^3 \ln \alpha = 62$) agreed much better with experimental values ($10^3 \ln \alpha = 73$) and correctly reproduced the inversion observed in the experimental fractionation (Figure 4, green line), albeit predicting the inversion to occur at a temperature that was too low. Including some exact exchange in the PBE functional (PBE0) improved the overestimation of NQEs in the O–H stretching coordinate and gave the correct qualitative behavior at 300 K ($10^3 \ln \alpha = 90$). It was also shown that the magnitude of the fractionation ratio predicted was highly correlated with the number of “proton excursions” observed in path-integral simulations of the functional.

In addition, to assess the ability of DFT to reproduce hydrogen-bond structures when NQEs are included, another recent study benchmarked the energy of configurations in bulk water obtained from a variety of DFT exchange-correlation functionals against quantum Monte Carlo, which provides a highly accurate benchmark.¹⁶² When the configurations were obtained from a PIMD simulation, GGA functionals gave energy errors about twice as large as for hybrids. However, when the configurations were taken from a classical MD simulation, much smaller differences were observed between GGAs and hybrids. This is because quantum fluctuations allow the O–H atoms in a covalent bond to sometimes come closer, and GGAs tend to predict these bonds to be too soft in water.^{163,164} Because of the local nature of this problem, adding exact exchange¹⁶⁴ or a monomer correction to DFT¹⁶⁵ can

somewhat alleviate this problem of the internal O–H covalent bonds in the water molecule.

3.3. Water at Interfaces

The quantum nature of water at interfaces and under confinement has been examined in a number of recent simulation studies (see, e.g., refs 109–111, 139, and 166–171), sometimes also with a view toward understanding related experiments. These efforts have started to reveal evidence under certain circumstances for quite strong quantum effects including tunneling, proton delocalization, and hydrogen-bond symmetrization. We do not attempt a comprehensive review of the literature in this very interesting area, but rather focus now on one particular area where strong evidence for such effects has been obtained. Specifically, we consider the adsorption of water on metal surfaces. Various experiments and simulations^{107,108} provide evidence to suggest that, in some adsorption systems, rather short O–O separations can be obtained. This comes about because water molecules on metal surfaces generally bind preferentially above individual metal atoms of the substrate;¹⁷² for close-packed surfaces of metals with relatively small lattice constants, molecules at adjacent sites can therefore be in close proximity. On the (111) surface of Ni, for example, this can, in principle, lead to O–O separations as short as 2.5 Å. With this in mind, Li et al.¹⁰⁹ performed a series of ab initio PIMD simulations for water overlayers on several metal surfaces [Ni(111), Ru(0001), and Pt(111)]. The particular overlayers considered were mixtures of water molecules and hydroxyl groups that form on many metal surfaces, for example, when water and oxygen are coadsorbed.¹⁰⁸ Figure 6 reports some of the key results of that study for the structures of the overlayers formed on the three surfaces. Specifically, distributions of O–H and O–O bond



lengths show that, as one goes from the substrate with the largest lattice constant considered (Pt) to that with the smallest (Ni), the degree of hydrogen-bond symmetrization increases. In some situations on Pt and Ru, HO–H–OH-like complexes are observed (Figure 6g), whereas on Ni, protons are almost perfectly shared between neighboring molecules (Figure 6h).

Direct experimental evidence for hydrogen-bond symmetrization has yet to be obtained for the specific overlayers probed in ref 109. However, evidence for a very closely related adsorption structure comprising a single H₂O and a single OH was obtained by means of STM on the (110) surface of copper.¹¹⁰ In an elegant set of highly controlled experiments, Kumagai et al. managed to observe the formation of a water–hydroxyl complex from individual coadsorbed water molecules and hydroxyl groups. The complex that formed had a symmetric oval shape, and along with complementary DFT calculations, it was argued that the oval image was evidence for the formation of a HO–H–OH-like complex with the H between the two oxygens shared symmetrically. In this particular case, the ZPE level along the hydrogen bond sits quite a bit above the very small potential energy barrier for proton transfer along the O–O axis. Kumagai et al. have also performed a number of similar studies of related adsorption systems, and much of this work was reviewed recently.¹⁷³ Among other things, their work revealed clear experimental evidence of hydrogen-bond exchange in adsorbed water clusters,¹⁶⁹ in support of earlier theoretical predictions.¹⁶⁷

3.4. Simple Diabatic Model of NQEs in the Hydrogen Bond

As discussed, many properties suggest that water becomes less structured with increasing NQEs upon replacing D with H. However, other properties, particularly those at higher temperatures, suggest the opposite trend. The concept of competing quantum effects provides an organizing principle to understand this behavior, as well as a range of other phenomena in hydrogen-bonded systems.^{4,7,9} Figure 1 shows two hydrogen-bonded water molecules and highlights the vibrational modes that are of particular relevance for a single O–H bond: the stretch, a bend in the plane of the water molecule, and a bend out of the plane of the water molecule. A key property of hydrogen bonding is that the frequencies of these modes vary significantly with the distance *R* between the oxygen atoms. Furthermore, as *R* increases, the hydrogen bond between the molecules becomes weaker, causing the stretching modes to stiffen and the bending modes to soften.⁴

The physics of these competing quantum effects is captured within the potential energy surface (PES) given by a recently introduced simple model for hydrogen bonding based on two electronic diabatic states.⁵ The two states differ by transfer of a proton, and the model is similar in spirit to empirical valence bond models.^{174–178} It uses only two free parameters that describe the energy and length scale associated with the coupling of the two diabatic states. The key descriptors are the proton-donor–acceptor distance *R* (i.e., the separation between the oxygen atoms) and the difference ϵ between the proton affinities (or, equivalently, pK_a values) of the donor and the acceptor. The model gives a ground-state PES, $E(R, r, \varphi)$, where *r* is the length of the O–H bond in the donor molecule and φ is the deviation of the hydrogen bond from linearity (as depicted in Figure 1). If one uses this surface to calculate the classical (harmonic) stretching and bending vibrational frequencies, one finds that, as *R* increases, the stretching frequency increases, and the bending frequency decreases, with associated changes

of their ZPE in opposite directions. This competition between the stretch ZPE, which favors stronger and shorter hydrogen bonds, and the bend ZPE, which favors longer and weaker hydrogen bonds, is the essence of competing quantum effects in hydrogen-bonded systems.

Despite its seeming simplicity, this model of the hydrogen bond and its competing quantum effects describe a wide range of experimental observations, from bond lengths to vibrational frequencies, including both primary and secondary isotope effects. For example, the observed correlation with *R* of both the O–H stretching and the out-of-plane bending frequencies, for a wide range of compounds, is quantitatively described by this model.⁴ Indeed, in going from the gas phase to liquid water, the OH stretching frequency is measured experimentally to decrease from 3657 to 3404 cm^{−1}.^{51,52} In contrast, the OH (in-plane) bending frequency increases from 1595 to 1644 cm^{−1}. This clearly shows competing quantum effects: Hydrogen bonding decreases the ZPE associated with stretching and increases that of the bending vibrations. This has also been seen in high-pressure experiments on ice VIII: With increasing pressure, the frequency of the stretching modes decreases, and those of the librational modes increase.^{179,180} In addition, as shown in Figure 2b, this model also captures the dependence of the H/D secondary geometric isotope effect on the O–O distance *R* for a wide range of compounds.

4. METHODS TO SIMULATE NUCLEAR QUANTUM EFFECTS

NQEs must be included in any atomistic simulation that aims at capturing the whole spectrum of properties of systems that contain light atoms. In molecules or in solids, they can often be modeled accurately using a harmonic or quasiharmonic approximation. However, in liquids such as water, it is necessary to fully account for anharmonic and entropic effects, requiring a more comprehensive treatment of NQEs.

One of the most promising approaches to include NQEs in the static equilibrium properties of anharmonic systems are techniques based on the Feynman path-integral formulation of quantum mechanics,^{181,182} which can be used in conjunction with Monte Carlo^{183–185} or molecular dynamics^{186–189} to treat exactly the quantum statistics of an ensemble of distinguishable nuclei. It will be evident already from the discussions in section 3 that these techniques have found widespread use; mainly, this is due to the combination of effective algorithms that have reduced their computational cost combined with increases in computational power. In section 4.1, we provide a brief introduction to path-integral methods along with specific comments and recommendations about their application to liquid water. We only consider distinguishable particles because nuclear exchange is expected to play a minor role in the description of liquid water because the two spin states, ortho and para water, are closely degenerate in gas phase, which we expect to carry over also to the condensed phase.¹⁹⁰ For a detailed discussion of path integrals for bosons and fermions, we direct the reader to review articles focusing on that topic.^{183,191}

Next, we address the techniques that can be used to approximate the effects of NQEs on time-dependent properties. Because of the challenge of real-time quantum dynamics, which is formally exponentially scaling in both time and dimensionality, currently no method exists that allows one to obtain exact results for complex condensed-phase systems such as water. Nonetheless, significant progress has been achieved in the area

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of semiclassical quantum dynamical methods in the past decade and a discussion of these approaches will be presented in section 4.2.

4.1. Simulating Quantum Static Properties

Imaginary-time path-integral (PI) methods represent the most convenient framework to introduce NQEs in atomistic simulations to obtain static equilibrium properties. The path-integral representation is based on the general expression for the expectation value of an operator:

$$\langle \hat{O} \rangle = \int d\vec{R} \langle \vec{R} | \hat{\rho} | \vec{R} \rangle O(\vec{R})$$

where $\hat{\rho}$ is the density operator which, for simplicity, we assume to be diagonal in configuration space. In the case of the canonical ensemble, which we consider here (generalization to other ensembles is straightforward), the density operator takes the form $\hat{\rho} = e^{-\beta \hat{H}}$. Using the simple property that the product of density matrices is a density matrix, $e^{-(\beta_1 + \beta_2)\hat{H}} = e^{-\beta_1 \hat{H}} e^{-\beta_2 \hat{H}}$, the expectation value can be rewritten as a path integral:

$$\langle \hat{O} \rangle = \int \left[\prod_{i=1}^P d\vec{R}_i \langle \vec{R}_i | e^{-\tau \hat{H}} | \vec{R}_{i+1} \rangle \right] \frac{1}{P} \sum_i O(\vec{R}_i)$$

where $\tau = \beta/P$. This expression is exact for any value of P ; it merely replaces the density matrix at an inverse temperature β with a product of P density matrices at the smaller inverse temperature τ . Because there are no known analytic expressions for the density matrix of an arbitrary Hamiltonian at finite inverse temperature, we must resort to approximate expressions in practical calculations. In almost all cases, the approximations employed are exact in the limit $\tau \rightarrow 0$, which means that computed quantum averages are exact in the limit of infinite P and convergence to this limit must be carefully assessed in all calculations. Approximate expressions typically offer a trade-off between the speed of convergence with P and the complexity of the expression (in terms of computation, ease of implementation, robustness, etc.). In the following, we discuss the main alternatives used in practical calculations and their relevance to the study of liquid water. In addition, we try to offer useful advice about their use when possible.

4.1.1. Second-Order Approximation. Also known as the primitive approximation, the second-order approximation is the simplest and most common approximation to the short-time thermal density matrix. It is based on the Trotter expansion¹⁹²

$$e^{-\tau \hat{H}} = [e^{-(\tau/2)\hat{V}} e^{-\tau \hat{T}} e^{-(\tau/2)\hat{V}}] e^{O(\tau^2)}$$

and is correct to order τ^2 . It is particularly convenient, as it leads to a classical Hamiltonian that is isomorphic to a “ring polymer” composed of P copies of the system, with corresponding atoms connected by harmonic springs, as shown in Figure 7. P is referred to as the number of polymer beads. This form allows the partition function to be sampled by Monte Carlo (PIMC) or molecular dynamics (PIMD), using similar algorithms that can be used for the corresponding classical simulation. Although this approach requires only the evaluation of energies and forces for the P replicas and properties converge to the quantum limit with increasing P , the convergence is rather slow. In the case of water under ambient conditions, $P = 32$ is typically used for calculations with flexible water monomers, which is sufficient to obtain reasonably converged properties such as the quantum kinetic and potential energies and RDFs.^{6,17,152,193} Other quantities might converge

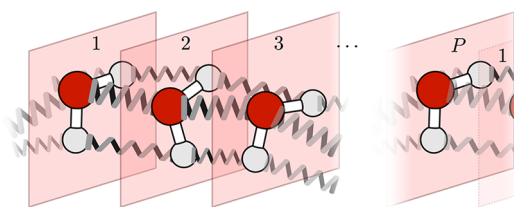


Figure 7. Representation of the imaginary-time path-integral isomorphism exploited in path-integral molecular dynamics (PIMD), ring-polymer molecular dynamics (RPMD), and centroid molecular dynamics (CMD). Each replica (bead) of the system is connected by harmonic springs to its adjacent replicas. Note that, because of the cyclic boundary conditions, the P th (last) ring-polymer replica is connected to the first one.

at a lower rate. For example, in the case of liquid water, convergence of the heat capacity to reasonable accuracy under ambient conditions typically requires $P \approx 128$.^{17,194,195} The second-order approximation is the most common implementation of PIMD and is the one employed in most software packages that offer PIMD as an option. However, as discussed below, in many cases, its efficiency can be enhanced by combining it with techniques such as reweighting, colored-noise thermostats, or ring-polymer contraction-based approaches.

4.1.2. Higher-Order Approximations. To improve the speed of convergence of path integrals with P , the use of higher-order approximations to Trotter's expansion has been investigated in detail over the past few decades. As discussed by several authors,^{196–199} the use of higher-order approximations (beyond second-order) invariably leads to nested commutators of the kinetic and potential energies, which, in turn, lead to complicated operators that are hard to evaluate in practice for general potentials. Even the simplest of these expansions, proposed by Takahashi and Imada,²⁰⁰ contains a term proportional to the square modulus of the force acting on the atoms. Although its use in PIMC methods is possible, its use in PIMD is limited because the derivative of this term in the Hamiltonian needed to generate the dynamics to sample the phase space requires calculation of the first-order derivatives of the force, that is, the Hessian matrix. Hence, for most potentials, any gain obtained by an improved convergence of properties with P is typically lost by the much more costly evaluation of the Hessian. For this reason, higher-order approximations to the thermal density matrix have been used only sporadically.

An interesting alternative to the use of high-order expansions in PIMD is the use of reweighted estimators.^{18,201,202} In this case, estimators derived from a high-order approximation are evaluated using a reweighting procedure with trajectories obtained from second-order PIMD calculations. This avoids the need for Hessian matrices if they do not appear in the estimators themselves, because the trajectories are propagated using the simple Trotter PIMD Hamiltonian. Although this scheme has been shown to lead to reduced efficiency as the system size grows, with the benefit becoming negligible at about 100 water molecules,²⁰³ it can be extremely useful for smaller systems when extremely accurate convergence of NQEs is required.

4.1.3. Accelerated Path Integrals with Generalized Langevin Equations. Generalized Langevin equations (GLEs) have recently been suggested as highly tunable thermostats in classical MD simulations^{204,205} and have been

Por que usar a matriz de densidade? Qual a vantagem em ver ela como uma multiplicação de várias matrizes densidade?

extended to nonequilibrium thermostat schemes that can be used to mimic nuclear quantum fluctuations in quasiharmonic systems.^{206,207} Furthermore, it has been shown that a combination of path-integral molecular dynamics and generalized Langevin equations (PI + GLE) can be conceived that yields virtually exact structural properties for harmonic systems with any number of replicas and is guaranteed to converge to Trotter path integrals in the limit of large P ¹⁵ (because the effect of the nonequilibrium GLE disappears in this limit, leaving the standard second-order expression). Because the GLE term effectively guarantees convergence of the harmonic terms, only anharmonicities need to be recovered directly by increasing the number of beads, leading to a dramatic reduction in the number of replicas required for convergence. This idea was later extended to also provide accurate estimates for momentum-dependent observables, such as the quantum kinetic energy (PIGLET).¹⁶ For example, as shown in Figure 8, for liquid water under ambient conditions, PIGLET with $P =$

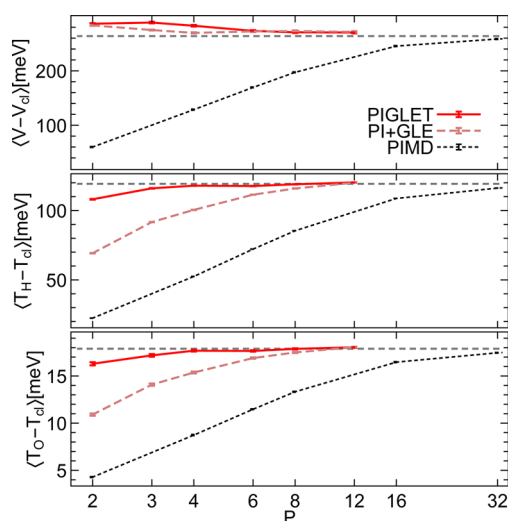


Figure 8. Quantum contributions to the potential energy and to the kinetic energies of hydrogen and oxygen atoms as computed by the centroid virial estimator for a simulation of a flexible water model⁶ at $T = 298$ K, plotted as a function of the number of beads. Note the much accelerated convergence of the kinetic energy when using PIGLET compared to PI + GLE. Adapted with permission from ref 16. Copyright 2012 American Physical Society.

6 already leads to well-converged properties. The additional computation necessary to implement the PIGLET thermostat is small compared to the evaluation of the forces in most systems, leading to a negligible overhead. PIGLET is thus currently one of the most effective approaches for incorporating NQEs in the calculation of static properties.

4.1.4. Ring-Polymer Contraction. Another alternative to accelerate the cost of converging PIMD calculations is the use of the ring-polymer contraction (RPC) approach.^{13–15} The main idea of RPC, which shares some similarities with the ideas behind multiple-time-step algorithms in classical MD simulations,²⁰⁸ is to break up the interaction into stiff and soft components and use the Trotter expansion in such a way that the stiff components, such as chemical bonds, are evaluated with a smaller value of τ . This can be easily seen in the following partitioning of the density matrix

$$e^{-\tau\hat{H}} \approx e^{-(\tau/2)\hat{V}_{\text{soft}}} \left[e^{-(\tau/2n)\hat{V}_{\text{stiff}}} e^{-(\tau/n)\hat{T}} e^{-(\tau/2n)\hat{V}_{\text{stiff}}} \right]^n e^{-(\tau/2)\hat{V}_{\text{soft}}}$$

In this case, the stiff interaction is evaluated n times more than the soft interaction. Because the soft component is slowly varying in space compared to the stiff component, it can then be evaluated on a contracted ring polymer, consisting of fewer beads $P' = P/n$, with negligible loss in accuracy. If the stiff component is chosen such that it can be evaluated much more quickly than the soft component, which is evaluated only P' times, then one can substantially decrease the cost of the calculation. One of the main benefits of this idea is that it can be combined with almost any other PIMD approach, including those presented above, leading to dramatic improvements in the efficiency of path-integral simulations. The main challenge is the need to divide the interaction into stiff and soft contributions. In the case of liquid water under ambient conditions, using an empirical water potential,⁶ RPC makes it possible to evaluate the intramolecular part of the potential on the full $P = 32$ ring polymer, whereas the expensive long-range electrostatic interactions can be computed just once, on the centroid.^{14,15} RPC approaches are thus highly appealing to use whenever possible and have been implemented in the i-PI program²⁰⁹ and the graphical-processing-unit- (GPU-) accelerated OpenMM code.^{210,211} Owing to the need to separate the time scales present in the potential energy, the application of RPC to ab initio PIMD simulations has provided more of a challenge. However, recent work has shown that RPC can be used to dramatically accelerate both DFT¹⁵⁵ and MP2²¹² calculations, while retaining chemical reactivity, by using a reference potential to create a smooth difference force that can then be evaluated on the contracted ring polymer.

4.1.5. Other Approximations. The approximations discussed above represent formally exact methods in the limit of infinite P . As long as one is careful in testing convergence with P , exact quantum averages can be obtained for any interaction type and strength. Other methods exist that lead to approximate results but typically at reduced computational cost and complexity. The simplest of these approaches is the use of perturbation theory to obtain quantum corrections to classical averages such as the Wigner–Kirkwood formula.^{213,214} It is worth noting that many of these approaches can be easily derived from the path-integral formulation and are associated with the reweighted estimators discussed above when used in combination with classical simulations instead of second-order expansions. Another closely related approach is the use of Feynman–Hibbs effective potentials and similar ways to incorporate NQEs approximately through renormalized interactions.^{181,215} These schemes are typically based on the use of high-order expansions for the thermal density matrix to construct corrections to the interactions that can be used in an otherwise classical simulation. A very recently introduced approach along these lines is to use the expansion of the free energy in orders of $\hbar$ (Planck’s constant) to obtain a correction to the second-order path-integral partition function for a given P .²¹⁶ This perturbed path-integral approach was shown to accelerate the convergence of the energy of liquid water, with convergence achieved at $P = 6$ compared to the usual value of $P = 32$.

It is also worth mentioning that a number of studies have attempted to account for NQEs in water by increasing the temperature of an otherwise classical simulation, for example, the use of classical results at higher temperatures as a means to reproduce quantum results at $T = 300$ K.²¹⁷ This procedure can be seen as a “poor man’s” version of the most primitive form of a quantum GLE (section 4.1.3), which is based on the idea of

simulating ZPE by enforcing a frequency-dependent effective temperature on different normal modes.²⁰⁶ However, the use of a uniform, modified classical temperature misses the strong anisotropy of the quantum momentum distribution, and can only reproduce a subset of the quantum properties. Furthermore, in hydrogen-bonded systems, such as water, that exhibit strong competing quantum effects, even the properties that appear correct might be reproduced for the wrong reason and will be strongly dependent on the model chosen to describe interatomic forces. For example, the temperature shift required to replicate NQEs for the single property of the change in the first peak of the O–O RDF of room-temperature liquid water was shown to range from 5 to 40 K depending on the water model chosen,¹⁰² and one would expect different DFT exchange-correlation functionals to show similar variability.²⁰ Hence, even for a single property, the effective temperature change needed to account for NQEs can vary wildly.²¹⁸

Although these more ad hoc approaches eliminate altogether the need for multiple replicas and are therefore as inexpensive as a classical simulation, they do so at the expense of introducing uncertainties that are difficult to quantify. This is particularly worrisome in a system such as water that involves significant molecular anisotropy and a competition of NQEs along different molecular axes. Considering the availability of efficient and systematically converging techniques such as PIGLET and RPC, we expect that the need to use such approximate approaches will be much decreased in the future.

4.2. Simulating Quantum Dynamical Properties

When it comes to including NQEs in simulations of dynamical (time-dependent) properties, the currently tractable computational techniques are much less reliable. In the case of gas-phase aqueous systems and small clusters, high-end benchmark calculations exist^{219,220} that demonstrate the dramatic effects of quantum coherence on the very low-temperature vibrational spectrum of hydrogen-bonded systems. Unfortunately, because the cost of these exact methods grows exponentially with the dimensionality of the problem, one has to resort to approximate methods to treat the condensed phases of water. In the study of chemical reactions at very low temperature, quantum instanton techniques²²¹ provide a robust framework for including ZPE and tunneling effects on reaction rates, but they are not easily extended to room temperature and the liquid state. In practice, the techniques that have been used more often to approximately describe quantum dynamics in water can be roughly classified in two different classes.

One class of methods treat a subset of the degrees of freedom quantum mechanically, embedding this quantum subsystem into the bulk environment, which is treated as static or as evolving in time according to classical dynamics. This family of techniques includes those based on the semiclassical theory of line shapes²²² that have been used with great success to model the response of an isolated chromophore in water, comparing successfully with experimental techniques ranging from simple infrared absorption²²³ to more sophisticated spectroscopies.²²⁴ Another possibility is to use the local monomer (LMon) approximation, which is based on selecting a subset of the local normal modes of one water molecule, computing the variations of the potential energy and the dipole moment as a function of the distortion of the molecule along these coordinates, and solving the Schrödinger equation for these modes in a fixed electrostatic environment.^{225,226}

The other class of methods treat the whole system on equal footing but heavily approximate the underlying nuclear dynamics. These methods typically utilize imaginary-time PIMD trajectories (section 4.1) to construct initial configurations consistent with quantum statistics, from which approximate trajectories are then initiated to calculate the relevant time correlation function for the dynamical property of interest. Linearized semiclassical initial-value representation methods use Newtonian dynamics to evolve these trajectories in time,^{227–230} which, in the case of liquid water, has been shown to be seriously affected by issues arising from unphysical ZPE leakage.²³¹ Very recently, a modified formulation of this approach, Matsubara dynamics, was introduced that preserves detailed balance and might help formulate more rigorous approximations of quantum dynamics.^{232,233}

Another possibility involves the use of modified forms of PIMD to generate approximate real-time correlation functions. Ring-polymer molecular dynamics (RPMD) simply evolves the ring polymer according to the discretized path-integral Hamiltonian,^{11,234–236} whereas centroid molecular dynamics (CMD) approximates the centroid-density dynamics by evolving the center of mass of the ring polymer over the potential of mean force obtained by averaging over the internal fluctuations of the ring polymer.^{237–239} Both techniques can be shown to yield exact quantum dynamics in some simple limiting cases, and some progress has been made toward giving formal justification to their use.^{232,240,241} In general, they can be expected to be more reliable for the evaluation of quantum corrections to long-time dynamical properties (e.g., diffusion coefficients). CMD and RPMD have been used to study diffusion and rotational relaxation in neat water,^{6,152,242} and CMD has been applied together with a multistate empirical valence-bond potential to study the dynamics of the excess proton in bulk water.¹²² However, both methods have been shown to exhibit serious artifacts when used to compute properties that are related to nonlinear operators and high-frequency vibrations.^{243–250} The latter problem can be mitigated by attaching thermostats to the internal modes of the ring polymer, a technique that can be seen as being halfway between RPMD and CMD.²⁵¹ In fact, one can see that, as long as one avoids stepping outside each method's comfort zone, CMD and thermostated RPMD give results that are consistent with each other and with methods based on a quantum description of a subset of the degrees of freedom.²⁵¹ NQEs are indispensable if one wants to achieve a quantitatively accurate description of the dynamical properties of water, but the differences between the various methods are comparable to those one should expect from performing simulations on an imperfect potential energy surface.

4.3. Computational Infrastructure and Codes to Include Nuclear Quantum Effects

One of the aspects that hinder more widespread modeling of NQEs in simulations of water arises from the technical complexities of performing a path-integral simulation—particularly in conjunction with a first-principles solution of the electronic structure. A few major codes for atomistic simulations provide some implementation of PIMD,^{210,211,252–258} but often, these implementations do not cover the whole spectrum of techniques discussed in this review or are not very actively maintained. A possible solution to this issue could come from a more modular design of the codes. A very convenient feature of PIMD is that the technicalities of the

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implementation do not involve a tight coupling with the internal workings of how interatomic forces within each replica are computed. Hence, it is possible to **separate the details of path-integral methods from those of the force calculation** and to develop a library of full-fledged PIMD routines that simply rely on one of the existing atomistic simulation packages to compute the energy and forces for a given configuration of the atoms. **i-PI** is a recently developed package that implements this idea in a radical way, using interprocess communication and a client–server design, so that any electronic structure or force field code can be easily patched to act as an energy and force back end to quantum simulations of nuclear statistics and dynamics.²⁰⁹ The same idea could be realized in a more closely integrated manner whenever the force back end provides a standardized library interface.

5. OUTLOOK AND ONGOING CHALLENGES

The past decade has seen some significant advances in the characterization and understanding of NQEs in water. The challenge of the next decade will be to **exploit these advances to provide a complete picture of water's NQEs in all regions of its phase diagram, particularly the deeply supercooled regime where these effects might be expected to be particularly important, as well as in its solvation properties and in broken-symmetry environments such as interfaces and in confinement.** These environments can require longer time scales and larger system sizes to simulate and have thus proved elusive, but with recent developments, they are now within reach. It will be interesting to see how the competition between quantum effects in water is mediated by such environments. Finally, below, we highlight three areas that we believe are essential to develop to obtain a more complete understanding of NQEs in water.

5.1. Experiments

DINS has revealed the momentum distribution of protons and can be compared to path-integral simulations.^{19,32,36,67,149,151} Furthermore, it has illuminated competing quantum effects.^{4,6,9,32} However, there remains some ambiguity associated with interpreting DINS experiments. The process of going from the data to robust (i.e., noncontroversial) spatial probability distributions for protons, particularly ones involving proton delocalization, needs to be improved.

In addition, new experiments that have probed isotope effects in deeply supercooled water have shown divergences between the static length scales in D₂O and H₂O, as well as a series of other intriguing observations.^{37,91} Given that a number of quantum effects have been demonstrated in the dynamics of other glassy quantum systems arising from quantum ZPE “swelling”, it will be interesting to see if these effects play a role in supercooled water.^{259,260}

5.2. Potential Energy Surfaces

Path-integral simulations will only be as good as the potential energy surfaces that they use. NQEs such as tunneling are particularly sensitive to the energy barrier for proton transfer. For example, we described work that showed how calculated isotope effects vary significantly with the DFT exchange-correlation functional that is used. This is because the potential energy surface, particularly with respect to the proton-transfer coordinate, is quite sensitive to the oxygen–oxygen separation and to the level of quantum chemical theory. For example, for H₃O₂[−], the PBE functional predicts a shared symmetric proton even in classical MD simulations, whereas more accurate

simulations using second-order Møller–Plesset perturbation theory (MP2) or the BLYP functional show that the protons actually exhibit a double-well potential.⁹ This difference becomes particularly important for properties that are determined by rare events (i.e., thermal and quantum fluctuations to short hydrogen bonds). The development of new DFT exchange-correlation functionals and the ability to use hybrid functionals^{217,261,262} with vdW corrections^{163,263–265} offers one such opportunity.²⁶⁶ In addition, recent developments in electronic structure theory and ongoing increases in computational power have made simulations using MP2 theory possible for condensed-phase water,²⁶⁷ and recent algorithms have also improved its scaling.²⁶⁸ For small gas-phase molecules, higher-level methods, such as coupled cluster theory, can also be utilized with PIMD simulations.²⁶⁹ Furthermore, recent simulations of liquid water using highly accurate quantum Monte Carlo surfaces suggest that PIMD simulations of this kind might soon become practical.²⁷⁰ New accurate intermolecular potential energy surfaces, such as MB-pol, also present new opportunities for more accurately identifying the role of NQEs in pure water.^{29,271–273}

5.3. Methods and Software for Efficiently Including Nuclear Quantum Effects

The combination of advanced path-integral algorithms with ongoing significant increases in computational power and the exploitation of novel computational architectures such as GPUs^{75,274–276} now allow ab initio PIMD simulations on condensed-phase aqueous systems of hundreds of atoms to be performed on a fairly routine basis. This, combined with the wider availability of codes to perform such calculations, will extend the ability to obtain much more detailed sampling and to reach the system sizes needed to simulate liquid water at interfaces with other materials. Owing to the larger range of hydrogen-bond strengths and, hence, modulation of the ZPE in these environments, one might expect to see pronounced NQEs. These developments will also allow simulations to reach the time scales needed to start to probe the role of quantum effects in more slowly relaxing environment where rare events, which are often greatly enhanced by NQEs, give rise to relaxation.

The ability to treat the quantum *dynamics* of aqueous system, however, still remains elusive, and although methods such as CMD and RPMD can be easily applied to aqueous systems and appear to give reliable results for properties such as diffusion constants, rates, and reorientation times, their performance on infrared spectra leaves cause for concern. It will be interesting to see if the new generation of ensemble-conserving linearized path-integral methods such as the Feynman–Kleinert quasi-classical Wigner method,²⁷⁷ path-integral Liouville dynamics,²⁷⁸ or Matsubara dynamics²³³ will yield more accurate results for such properties when directly applied to water or if they might yield directions for improving existing methods such as RPMD and CMD.

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The authors declare no competing financial interest.

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Ross H. McKenzie is a Professor of Physics at The University of Queensland in Brisbane, Australia. He was an undergraduate at the Australian National University and received a Ph.D. from Princeton University in 1989. Ross' research involves using quantum many-body theory to understand the properties of complex materials, ranging from fluorescent proteins to molecular superconductors. He is author of the blog "Condensed Concepts: Ruminations on Emergent Phenomena in Condensed Phases of Matter" (condensedconcepts.blogspot.com). Co-authorship of this review continues a family tradition: In 1978 and 1983, Ross' father, Hugh A. McKenzie, together with John T. Edsall, wrote reviews about water and its interaction with electrolytes and proteins.

Angelos Michaelides obtained a Ph.D. in Theoretical Chemistry in 2000 from The Queen's University of Belfast. After this, he worked as a postdoctoral research associate and junior research fellow at the University of Cambridge and then at the Fritz Haber Institute in Berlin, Germany, as an Alexander von Humboldt Research Fellow. Subsequently, he was promoted to staff scientist and research group leader at the Fritz Haber Institute. In 2006, he moved to University College London, where, since 2009, he has been Professor of Theoretical Chemistry. Research in his group (www.chem.ucl.ac.uk/ice) involves computer simulations of catalytic and environmental interfaces, aiming at reaching a fundamental new understanding of

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Thomas E. Markland received his D.Phil. in Chemistry from Balliol College, University of Oxford, in 2009 under the supervision of Professor David E. Manolopoulos. After postdoctoral work with Professor Bruce J. Berne at Columbia University, he moved to Stanford University in 2011 as an Assistant Professor in the Department of Chemistry. His research interests lie broadly in the development of theoretical and simulation methods and their application to explain effects observed in the classical and quantum dynamics of chemical systems. He has introduced numerous algorithms to improve the efficiency of quantum path-integral approaches and was an early developer of the ring-polymer molecular dynamics method. He has applied these developments to elucidate the role of quantum mechanics in hydrogen-bonded systems ranging from water to enzymatic proton networks.

ACKNOWLEDGMENTS

We thank Lars G. M. Pettersson, Lu Wang, and Ondrej Marsalek for providing insightful comments and suggestions for this review. This review was initiated during the Nordita (Nordic Institute for Theoretical Physics) scientific program "Water - the Most Anomalous Liquid". Additional financial support for this program was provided by the Royal Swedish Academy of Sciences through its Nobel Institutes for Physics and Chemistry, the Swedish Research Council, and the Department of Physics at Stockholm University. M.C. acknowledges funding from CCMX and the Swiss National Science Foundation (Project 200021-159896). P.G.K. is grateful for the financial support of the Natural Sciences and Engineering Research Council of Canada. A.M. and W.F.'s work was supported by the European Research Council under the European Union's Seventh Framework Programme (FP/2007-2013)/ERC Grant Agreement 616121 (HeteroIce Project). A.M. was also supported by the Royal Society through a Wolfson Research Merit Award. M.A.M. was supported by the Predictive Theory and Modeling for Materials and Chemical Science program of the U.S. Department of Energy (U.S. DOE), Office of Science, Office of Basic Energy Sciences, and by the U.S. DOE at Lawrence Livermore National Laboratory under Contract DE-AC52-07NA27344. T.E.M. was supported by the U.S. DOE, Office of Science, Office of Basic Energy Sciences, under Award DE-SC0014437. T.E.M. also acknowledges support from a Cottrell Scholarship from the Research Corporation for Science Advancement and an Alfred P. Sloan Research fellowship.

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